arXiv:2312.10794v5 [cs.LG] 21 Aug 2025

A MATHEMATICAL PERSPECTIVE ON TRANSFORMERS

BORJAN GESHKOVSKI, CYRIL LETROUIT, YURY POLYANSKIY,
AND PHILIPPE RIGOLLET

Abstract. Transformers play a central role in the inner workings of large language models. We develop a mathematical framework for analyzing Transformers based on their interpretation as interacting particle systems, with a particular emphasis on long-time clustering behavior. Our study explores the underlying theory and offers new perspectives for mathematicians as well as computer scientists.

Contents

1. Outline 2

Part 1. Modeling 4
2. Interacting particle system 4
3. Measure to measure flow map 9

Part 2. Clustering 17
4. A single cluster for small β 18
5. A single cluster for large β 20
6. The high-dimensional case 20

Part 3. Further questions 27
7. Dynamics on the circle 28
8. BBGKY hierarchy 31
9. General matrices 32
10. Approximation, control, training 36
Acknowledgments 37

Appendix 37
Appendix A. Proof of Theorem 4.1 37
Appendix B. Proof of Theorem 5.1 39
Appendix C. Proof of Theorem 4.3 42
Appendix D. Proof of Theorem 6.9 43
References 47

2020 *Mathematics Subject Classification.* Primary: 34D05, 34D06, 35Q83; Secondary: 52C17.
Key words and phrases. Transformers, self-attention, interacting particle systems, clustering, gradient flows.

1. Outline

The introduction of *Transformers* in 2017 by Vaswani et al. [VSP+17] marked a significant milestone in the development of neural network architectures. Central to this contribution is *self-attention*, a novel mechanism which distinguishes Transformers from traditional architectures, and which plays a substantial role in their superior practical performance. In fact, this innovation has been a key catalyst for the progress of artificial intelligence in areas such as computer vision and natural language processing, notably with the emergence of large language models. As a result, understanding the mechanisms by which Transformers, and especially self-attention, process data is a crucial yet largely uncharted research area.

A common characteristic of deep neural networks (DNNs) is their compositional nature: data is processed sequentially, layer by layer, resulting in a discrete-time dynamical system (we refer the reader to the textbook [GBC16] for a general introduction). This perspective has been successfully employed to model *residual neural networks*—see Section 2.1 for more details—as continuous-time dynamical systems called neural ordinary differential equations (neural ODEs) [CRBD18, E17, HR17]. In this context, an input $x(0) \in \mathbb{R}^d$, say an image, is evolving according to a given time-varying velocity field as $\dot{x}(t) = v_t(x(t))$ over some time interval $(0, T)$. As such, a DNN can be seen as a flow map $x(0) \mapsto x(T)$ from $\mathbb{R}^d$ to $\mathbb{R}^d$. Even within the restricted class of velocity fields $\{v_t\}_{t \geqslant 0}$ imposed by classical DNN architectures, such flow maps enjoy strong approximation properties as exemplified by a long line of work on these questions [LJ18, ZGUA20, LLS22, TG22, RBZ23, CLLS23].

Following [SABP22] and [VBC20], we observe that Transformers are in fact flow maps on $\mathcal{P}(\mathbb{R}^d)$, the space of probability measures over $\mathbb{R}^d$. To realize this flow map from measures to measures, Transformers evolve a *mean-field interacting particle system*. More specifically, every particle (called a *token* in this context) follows the flow of a vector field which depends on the empirical measure of all particles. In turn, the *continuity equation* governs the evolution of the empirical measure of particles, whose long-time behavior is of crucial interest. In this regard, our main observation is that particles tend to cluster under these dynamics. This phenomenon is of particular relevance in learning tasks such as *next-token prediction*, wherein one seeks to map a given input sequence (i.e., a sentence) of n tokens (i.e., words) onto a given next token. In this case, the output measure encodes the probability distribution of the next token, and its clustering indicates a small number of possible outcomes. Our results on a simplified but insightful toy model indicate that the limiting distribution is actually a point mass, leaving no room for diversity or randomness, which is at odds with practical observations. This apparent paradox is resolved by the existence of a long-time metastable state. As can be seen from Figures 3 and 5, the Transformer flow appears to possess two different time-scales: in a first phase, tokens quickly form a few clusters, while in a second (much slower) phase, through the process of pairwise merging of clusters, all tokens finally collapse to a single point. This appears to corroborate behavior observed empirically in trained Transformer models, which goes under the names *token uniformity*, *over-smoothing* [CZC+22, RZZD23, GWDW23, WAWJ24, WAW+24, DBK24, SWJS24], or *rank-collapse* [DCL21, FZH+22, NAB+22, JDB23, ZMZ+23, ZLL+23, NLL+24, BHK24, CNQG24]; see also Figure 1.

The goal of this manuscript is twofold. On the one hand, we aim to provide a general and accessible framework to study Transformers from a mathematical perspective. In particular, the structure of these interacting particle systems enables concrete connections to established mathematical topics, such as nonlinear transport equations, Wasserstein gradient flows, collective behavior models, and optimal configurations of points on spheres, among others. On the other hand, we describe several promising research directions with a particular focus on the long-time clustering phenomenon. The main results we present are new, and we also provide what we believe are interesting open problems throughout the paper.

The rest of the paper is arranged in three parts.

Part 1: Modeling. We define an idealized model of the Transformer architecture that captures two of the main characteristics of transformers: *self-attention* and *layer-normalization*. Following a perspective put forward in classical architectures such as ResNets [CRBD18, E17, HR17], we view the successive layers of a neural network as time discretizations of a dynamical system of interacting particles. Layer-normalization effectively constrains particles to evolve on the unit sphere $\mathbb{S}^{d-1}$, whereas self-attention is the particular nonlinear coupling of the particles done through the empirical measure (Section 2). In turn, the empirical measure evolves according to the continuity partial differential equation (Section 3). Even after significant simplifications, this toy model retains macroscopic characteristics of trained Transformers, namely clustering. We also introduce a simpler surrogate model for self-attention which has the convenient property of being a Wasserstein gradient flow [AGS05] for an energy functional that is well-studied in the context of optimal configurations of points on the sphere and sheds a complementary light of the source of clustering.

Part 2: Clustering. In this part we recall existing and establish new mathematical results that indicate clustering of tokens in the long-time limit. (See Figure 2 for a summary.) Our first results (Theorem 4.3 in Section 4 and Theorem 5.1 in Section 5) are in extreme regimes of a temperature parameter β^{-1} that appears in the equation. We then move to the high-dimensional case in Section 6, where we begin by recalling Theorem 6.1—a result of [MTG17], recently revisited in [CRMB24]—which entails long-time clustering at any temperature when $d \geqslant 3$. We provide an exponential rate of convergence in Theorem 6.3 when $d \geqslant n$—here n denotes the number of particles—. We complement this result with an even more precise characterization of the rate of contraction of particles into a cluster. Namely, we describe the histogram of all inter-particle distances, and the time at which all particles are already nearly clustered (Theorem 6.9).

Part 3: Further questions. We propose potential avenues for future research, largely in the form of open questions substantiated by numerical observations. We first focus on the case $d = 2$ (Section 7) and elicit a link to Kuramoto oscillators. We briefly show in Section 9.1 how a simple and natural modification of our model leads to non-trivial questions related to optimal configurations on the sphere. The remaining sections explore interacting particle systems that allow for parameter tuning of the Transformers architectures, a key feature of practical implementations.